

ABERDEEN DYCE, United Kingdom

WMO: 030910

Lat: 57.2050N

Lon: 2.2053W

Elev: 215

StdP: 14.58

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.0	(c) 27.6	(d) 20.5	(e) 15.5	(f) 27.8	(g) 23.0	(h) 17.5	(i) 31.4	(j) 34.7	(k) 43.1	(l) 31.7	(m) 42.5	(n) 2.8	(o) 0	(p) 0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	71.4	62.0	68.2	60.0	65.8	58.4	63.5	69.4	61.6	66.4	60.0	64.2	11.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.9	80.6	65.9	59.4	76.2	64.0	57.9	72.2	62.3	28.9	69.6	27.6	66.5	26.5	64.4	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 27.7	(b) 24.4	(c) 21.7	(e) 16.9	(f) 77.4	(g) 6.1	(h) 2.8	(i) 12.5	(j) 79.5	(k) 8.9	(l) 81.1	(m) 5.5	(n) 82.7	(o) 1.1	(p) 84.8		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			16.3	66.8	6.1	1.8	11.9	68.1	8.3	69.2	4.8	70.2	0.4	71.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	39.0	39.5	41.9	45.1	49.3	54.3	58.2	57.8	54.5	49.1	43.3	39.2
	DBStd	8.45	4.69	5.22	5.25	4.89	4.69	4.27	4.03	3.76	4.05	4.76	5.30	6.17
	HDD50	1756	340	296	257	159	69	10	0	1	8	74	207	335
	HDD65	6338	805	713	717	596	485	321	215	224	317	494	652	799
	CDD50	898	0	2	5	14	49	139	254	243	141	45	5	1
	CDD65	7	0	0	0	0	0	0	4	2	1	0	0	0
	CDH74	31	0	0	0	0	0	3	16	10	2	0	0	0
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0
	WSAvg	10.1	11.5	11.6	11.3	10.2	9.4	9.5	8.5	8.7	9.5	10.4	10.4	10.8
	PrecAvg	32.0	3.1	2.2	2.2	2.2	2.4	2.4	2.6	2.9	2.9	3.4	3.3	2.8
Precipitation	PrecMax	43.0	6.3	6.1	4.1	3.9	6.1	5.8	5.3	6.1	10.3	9.3	10.8	5.7
	PrecMin	20.3	0.5	0.3	0.4	0.6	0.5	0.7	0.5	0.5	0.8	0.4	0.8	0.7
	PrecStd	5.6	1.5	1.4	1.0	0.8	1.4	1.3	1.3	1.5	1.8	2.2	2.2	1.3
	PrecStd	5.6	1.5	1.4	1.0	0.8	1.4	1.3	1.3	1.5	1.8	2.2	2.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	57.8	60.3	63.9	68.3	72.6	76.0	74.4	72.0	63.8	58.4	55.8
		MCWB	49.3	50.6	50.2	54.3	57.9	62.5	65.1	64.0	62.2	57.6	52.9	51.1
	2%	DB	51.1	53.6	55.9	59.3	63.7	67.7	71.9	69.8	66.6	60.1	55.1	52.8
		MCWB	47.2	48.3	48.5	51.3	55.2	58.3	62.4	61.8	58.9	54.7	51.1	49.2
	5%	DB	48.6	50.3	52.9	56.0	60.4	64.7	68.2	66.8	63.6	57.7	53.0	50.2
		MCWB	45.2	45.7	46.7	49.3	53.2	56.8	60.6	59.3	56.8	53.5	49.7	47.1
	10%	DB	46.1	47.5	50.3	53.4	57.6	62.0	65.5	64.7	61.2	55.9	51.1	47.7
		MCWB	43.4	43.4	45.1	47.5	51.0	55.1	58.8	58.1	55.7	52.3	48.5	45.1
	0.4%	WB	50.1	51.5	51.9	56.3	59.5	63.3	66.6	65.4	64.0	58.4	53.9	51.9
		MCDB	53.1	56.4	57.8	62.5	66.2	71.3	74.5	73.0	69.9	62.5	56.4	55.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.7	48.7	49.5	52.5	56.5	60.0	63.8	63.0	60.3	56.1	52.1	49.7
		MCDB	50.6	53.0	54.7	57.7	62.6	66.2	69.9	68.1	65.1	59.0	54.1	52.4
	5%	WB	45.6	46.4	47.4	50.1	54.0	58.0	61.8	61.2	58.3	54.6	50.4	47.5
		MCDB	47.9	49.8	51.9	55.0	59.2	63.2	66.5	65.2	61.8	56.8	52.2	49.7
	10%	WB	43.8	44.0	45.5	48.2	52.0	56.2	60.2	59.6	56.8	53.2	48.9	45.5
		MCDB	45.9	46.9	49.8	52.5	56.4	60.7	64.4	63.0	59.8	55.3	50.8	47.6
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.9	11.2	11.9	12.8	11.7	12.0	12.2	11.8	10.4	9.8	9.2
		MCDBR	11.8	13.3	16.0	16.2	17.2	16.5	17.1	15.7	15.6	12.8	11.7	11.4
		MCWBR	10.2	10.0	10.7	10.3	10.5	10.0	10.2	9.4	9.9	9.6	10.0	10.0
	5% WB	MCDBR	11.0	12.6	15.0	15.5	16.1	15.5	15.2	14.4	13.6	11.5	10.7	10.7
		MCWBR	10.0	10.1	10.8	10.4	10.5	10.2	9.7	9.4	9.5	9.4	9.8	9.8
			10.0	10.1	10.8	10.4	10.5	10.2	9.7	9.4	9.5	9.4	9.8	9.8
Clear-Sky Solar Irradiance	taub		0.306	0.331	0.357	0.379	0.375	0.373	0.377	0.366	0.350	0.337	0.310	0.290
	taud		2.265	2.350	2.338	2.303	2.368	2.398	2.392	2.431	2.481	2.437	2.309	2.201
	Ebn at Noon		177	227	252	264	273	275	271	266	254	223	174	146
	Edh at Noon		16	23	29	35	35	35	34	31	25	20	15	13
All-Sky Solar Radiation	RadAvg		141	343	688	1138	1538	1514	1467	1167	805	422	183	97
	RadStd		18	32	76	109	82	122	114	77	65	37	16	12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page